Knowledge Representation & Reasoning

KRR is a part of AI that helps computer represent the knowledge about real world and use that knowledge to draw conclusions and make decisions

Knowledge is a information fact and understanding that the system has about a particular subject or a real world.

Knowledge representation:
Knowledge representation is a process of representing facts relationships in the structured ~~facts~~ form so that AI can use it for understanding and reasoning.

Reasoning: Reasoning is process of using existing knowledge facts, rules to drive new information or conclusion.

logic: logic is a method of representing knowledge using facts and rules and derive correct conclusion from them.

Types of logic
-> proposition logic
-> predicate logic
-> first order logic
-> fuzzy logic

1) Proposition logic:
->It is type of a logic in AI where knowledge is represented using propositions and each proposition either true or false
main components.

i) proposition: It is a statement that can either be true or false

ii) logical connectivity:
AND, OR, NOT, IFf, Implies.

3) Truth value
True or false (0 or 1)

iv) Truth Table: It is a table which show the every possible combination (input and output) for proposition.

Truth table for NOT gate

not (r)

A	¬A
0(F)	1(T)
0(F)	0(F)

Ex:- P: It is raining
¬A: It is not raining

AND(∧) gate. Truth table.

A	B	A∧B
0	0	0
0	1	0
0	0	0
1	1	1

Ex:- P: I have pen (T)
Q: I have a notebook (T)
conclusion: I have pen and notebooks (P∧Q).
note:- both statements must be true for conclusion to be true

Ex:- suppose a college says you need ID card and admit card to enter the exam
conclusion: both id card & admit card.

OR(∨)

A	B	A∨B
F	F	F
F	T	T
T	F	T
T	T	T

Ex:- P : I will go to home
Q : I will stay at college
conclusion: I will go to home or I will stay at college.

Note:- OR is true atleast one variable is true

Ex:- P: I will eat rice (T)
Q: I will eat dose (F)
P ∨ Q = I will eat rice or dose, (T)

Implies (→)
This implies shows if then relationship
Note:- It is false only when P is true but Q is false.

P	Q	P → Q
F	F	T
F	T	T
T	F	F
T	T	T

Ex:- P : It is raning (T)
Q : I carry an umbrella (F)
P → Q : If It is raining then I carry an umbrella (F)

Ex. P : I study well (T)
Q : I get good marks (T)
P → Q : If i study well then i get good marks (T)

Iff (⟷)
Iff (If and only if) it also called as by direction.

P	Q	P ⟷ Q
F	F	T
F	T	F
T	F	F
T	T	T

Ex:- P: person is a student (T)
Q: The person has student ID card (T)

P ⟷ Q : The person has student if and only if they have a student ID card (T)

Ex:- P: A number is divisible by 2 (T)
Q: The number is even (T)

P ⟷ Q : A number is divisible by 2 iff the number is even (T)

Ex. P: The password is correct (F)
Q: The user can login (T)
P ⟷ Q: The user can login iff the password is correct (F)

syntax on proposition logic
syntax here means rules for writing valid formulas in proposition logic

Basic elements
1) proposition symbol
2) logical connectivity
3) parenthesis.

semantic proposition logic
semantic means meaning or truth value of logical expressions.

syntax	semantics
Rules of forming expression syntax deals with the structure Ex: P ∧ Q P → Q	Truth of the expression sematics deals with the truth Ex: (P ∧ Q) is true.

Inference rule.

Inference means deriving new conclusion from known facts and rules.

Inference is used in reasoning from existing knowledge

Ex:- fact: It is raining then
rule: if it is raining then the road is wet.

conclusion: The road is wet.

Important inference rules.

1) Modus ponens

statement:- It is a inference rule used derive a conclusion from a implecation and a fact.

formula :- $\frac{p \to q \ (+) \quad p \ (+)}{\therefore q \ (+)}$

meaning: If p is true then Q is true
T → T p statement is true ∴ Q is true

note: The conforme p that is P is true then q is true then q is true

Ex:- P: ravi stadies
Q: ravi passes
p→q: If ravi stadies then he passes.
P: Ravi stadies
∴ Q: Ravi passes

Ex: P: The trafic is red
Q: The vehicals must stop
p→q: If the trafic is red then vechicals must stop
P: The trafic is red
Q: vechicals must stop

Ex: P: Battery is not charged
Q: The phone will not true on
p→q: If Battery is not charged then the phone will not turn on.
P: Battery is not charged
Q: the phone will not turn on.

Ex: P: It is raining.
Q: The road is not wet
p→q: If it is raining then the road is not wet
P: It is raining
∴ The road is not wet.

P: The door is locked
Q: The person can not enter
p→q: If the door is lock then the person can not enter
P: The door is lock
∴ The person can not enter

Modus tollens

It is a inference rule in proposition logic used to drive negation of a condition from negation of its result.

Note: It denies a that is q is false the p as to be false

. formula: $\frac{p \to q \quad \neg q}{\therefore \neg p}$

meaning: If p is true then q is true
q is false
∴ p is false.

Ex:-
P: It is raining
Q: The road is wet
$P \rightarrow Q$: If it is raining then the road is wet
$\neg Q$: road is not wet
∴ It is not raining.

Ex:- P: ravi studies
Q: ravi passes the exam
$P \rightarrow Q$: If ravi studies then ravi passes the exam
$\neg Q$: ravi not passes the exam
∴ ravi not studies.

Ex- P: student studies
Q: student does not fail
$P \rightarrow Q$: If student studies then students does not fail.
$\neg Q$: student fail.
$\neg P$: student not study.

→ Hypothetical Syllogism (HS)
It is a inference rule used to combain two conditional statements to derive new conditional conclusion.

syntax :-
Premise 1 : $P \rightarrow Q$
Premise 2 : $Q \rightarrow R$
Conclusion : $P \rightarrow R$.

Meaning
If P leads to Q and Q leads to R the conclusion is P leads R.

Ex:- premise 1 : If I study then I 'll pass in exam
premise 2: If I pass I'll get a bike.
conclusion : If i study then I'll get a bike.

Ex: premise 1 : If it rains ground becomes wet
premise 2 : If ground becomes wet it be slippery.
conclusion. If it rains then ground becomes slippery

→ Disjunctive syllogism
It is the inference rule where we have OR statement and one statement there to be false there four another statement is must be true.

syntax:
$P \vee Q$
$\neg P$
∴ Q.

meaning:
either P or Q is true, if P is false conclusion then Q must be true.

Ex:- P∨Q : I will go to college
¬P : I will not go to college
Q : I will go to movie.

Ex: P∨Q : either it is raining or it is suning
P = It is not raining
Q : It is suning.

Addition

It is a inference rule the allow us to take single true proposition and combain it with any other proposition using OR. The resulting must be true

syntax: P (T)
∴ P∨Q (T)

meaning:
If P is is true then P∨Q is true. Here Q can be any proposition.

Ex:- P : It is raining (T)
Q : It is suning.
∴ P∨Q : It is raining or suning (T)

P : Ravi is studing (T)
Q : Ravi is playing
∴ P∨Q : Ravi is studing or playing (T)

simplification.

It is a inference rule that allows us to take one part of the compound statement conected by AND and conclude that part alone

syntax: P∧Q (T) P∧Q (T)
∴ P (T) ∴ Q (T)

Ex:- P : Ravi is studing
Q : Ravi is listing to music
P∧Q : Ravi is studing and listing to music
P = Ravi is studing

Ex:- P : 10 is the even number
Q : 10 is greater than 5
P∧Q : 10 is the even number and greater than 5
P = 10 is the even number.

~~Conjoation~~. conjunction

It is a inference rule that allows to combain two separate true proposition into one compound proposition using AND

syntax:
P (T)
Q (T)
∴ P∧Q (T)

Ex:- P : computer is on (T)
Q : computer is conected to internet. (T)
∴ P∧Q : computer is on and conected to internet (T)

P : s is odd (T)
Q : s is greater than 3 (T)
∴ P∧Q = s is odd and greater than 3.

Resolution

It is a inference rule used to derived new conclusion combaining two clauses that contain proposition and its negation.

syntax:
premise 1 : P∨Q
premise 2 : ¬P∨R.
conclusion: Q∨R.

Ex:- P : Ravi will study R : Ravi will watch TV
Q : Ravi will play.
P∨Q : Ravi will study or play
¬P∨Q : Ravi will not study or play
Q∨Q : Ravi will play or watch TV

P : It is raining Q : I will carry on umbrella R : I will stay within home
P∨Q : It is raining or carry on umbrella
¬P∨Q = It is not raining or carry on umbrella
Q∨R : I will carry on umbrella or I will stay in home

Predicate logic / First Order

It is a formal system used to represents statement about objects and there properties or relationship using predicates. variables and quantifiers

- Simple meaning

It is simple tell us about what are we taking about

Ex:- statement:- Ravi is a student.

It can be written as student (Ravi)

↑ predicate ↓ subject/object.

another example

Statement: Ravi likes maths

It can be written as likes (Ravi, maths)

↓ predicate ↓ object / subject

predicate. → It is a function that represents a property of an object. or relationship between object.

Ex:- one variable

let P(x) = x is a student

if x = Ravi

P(Ravi) = x is a student.

∴ student (Ravi)

Ex:- Two variable.

let likes (x)(y) = x likes y

then, likes (Ravi, Maths)

↓ predicate

Teacher (x) → x is a teacher

Greater (5, 3) → 5 is greater than 3

quantifiers

It tell us how many object in a given domine statisfy the predicate

Two types of quantifiers

1) Universal
2) Existential

Universal quantifier (∀)

It is used when a statement is true for every object in a specify domain

syntax: ∀ → for all / for every / foreach.

∀ x P(x)

→ read it as for every x P(x) is true

Ex:- let Human (x) = x is a Human

Mortal (x) = x is a mortal.

all human are mortal Represent using universal Quantifier.

∀ Human (x) → mortal (x)

Every student studies.

studies (x) = x studies.

student (x) = x is a student

∀ student (x) → studies (x)

Every dog is a animal

dog (x) = x is dog

Animal (x) = x is a animal

∀ dog (x) → Animal (x)

read it as for every x if x is dog then x is a animal.

Every car as wheels

car (x) : x is car

wheel (x) : x has wheel.

∀x car(x) → wheel(x)

for every x, if x is car then x as a wheel

for every square as 4 sides

square (x) = x is square

as 4 sides (x) = x as 4 sides

∀x ; square (x) → has 4 sides (x)

every student likes every subject

→ Every student has a teacher.

$\forall x \forall y (student(x) \wedge Teacher(y)) \rightarrow Teaches(Teacher, students)$

→ Every mother loves her child

$\forall x \forall y (Mother(x) \wedge child(y)) \rightarrow loves(mother, child)$

→ All students likes all subjects.

$\forall x \forall y (Student(x) \wedge subject(y)) \rightarrow likes(student, subject)$

Extensial quantifier

There exist atleast one or some.

Example: Some student is intelligent.

student(x) = x is a student

intelligent(x) = x is a intelligent.

$\exists x [student(x) \wedge intelligent(x)]$.

→ Some students likes AI

$\exists x$ student(x) $\wedge$ likes (student, AI)

→ There is a teacher, who teaches Ravi.

$\exists x$ teacher(x) $\wedge$ teaches (teacher, Ravi)

→ There is a dog that is black

$\exists x$ dog(x) $\wedge$ black(x)

→ Ravi likes some student

$\exists x$ Ravi(x) $\wedge$ likes (Ravi, student)

Rule based system

It is a AI system that uses set of if than rules, to represents the knowledge and make the decisions

→ It is inference rule

Basic structure

if (condition) → Then (conclusion / action).

Ex:- if body (temperature is high) than patient has (fever)

(condition) (conclusion)

components of rule based system

1) knowledge based
2) inference ingin.
3) working memory.
4) user interface.

Knowledge base

It contains facts and rules. (if than)

inference ingin

apply rules to the avilable facts.

working memory

It stores current facts and processing information.

user interface.

allows user to enter the information. and recive the system result.

Ex:- A pateient as temperature 102° and cough. a system should determaind whether the patient as flue

→ knowledge base

fact temp = 102°

cough = yes.

Rules

R: if temp > 90° than fever

if fever + cough than flue.

→ inference ingin

Ex: temp = 102° rule 1 is statisfy. = fever

fever and cough. rule 2 is statisfy = flue.

→ working memory
It store the information

→ user interface
The patient as flue.

- forward chaining
It is a inference techique in AI that starts with known facts and applys rules step by step to derive new fact until a final conclusion is reached
→ It is called data deriven reasioning, because it starts from avilable data.

Ex:- student performance.
→ suppose AI system determaining whether student is eligible to shcolorship
mark-90
attendence = 95%

rules
if marks ≥ 80 than student is excellent.
if attendence >90% than attendance is good
if student is excellent and attendence is good than student is eligible for sholorship

forward chaining
step1 : start with known facts.
marks=90 attendence=95%
step2 : apply rule one: student is excellent.
step3 : apply rule two : attendence is good
step4 : apply rule three : student is eligible for scholorship

Backward chaining
It is a inference techinque in AI that starts with goal or consulsion, works backward throw the rules find the fact that support that goal.
→ It is also called goal deriven.
Ex-
step1 : is student is eligible for scholorship.
step2 : Find the rule the produce the goal.
rule three say if student is excellent and good at attendance than eligible for scholorship.
step3 : prove excellent. if marks ≥ 80 → excellent
marks = 90 ∴ 90 ≥ 80 → excellent
step4 : prove attendence is good. if all ≥ 90 → good att
attendence = 95%.
∴ 95 ≥ 90% → attendence is good
step5 : student is eligible to the scholereship

application of inference rules
→ medical digness. / checkup
→ frad detection.
→ recomendation system
→ falt diogney
→ robotics
→ expert system.

Ontology
It is formal representation of a knowledge in a particuler domine.
→ It tell us what things exists, what are there property how are related each.
Ex:- university.
Student
Teachers
course
subjects
} concepts.

Components of ontology

-> Concepts / class
It represents general catagries of things

-> Individuals
Theis are the specific object of the class.
Ex- student -> Akash
teacher -> some
course -> BCA

-> properities
It discrib the characteristic of an object.
Ex:- student -> name, age, Reg no, city.

-> relationship
How two concepts are individualy are ~~conne~~ conected.
Ex:- ! student studies course.
Akash studies BCA

-> rules:
It is also called axioms. this are the logical statements that define ~~de~~ driscution are the relationship between concepts.
Ex: Every teachers teaches at least one course.

advantages
-> knowledge represention systematicaly
-> knowledge sharing
-> knowledge reuse.
-> It can be help in drive new information
-> data integration -> combining data from different sources

applications
-> AI models
-> sematic webs
-> health care
-> e-commory
-> NLP

steps in ontology development.

Identify domain
↓
Identify concepts
↓
~~Identify individuals~~
define properties
↓
define Relationship → define rules
Create individuals
↓
use for reasoning

Identify domain
area of which knowledge is being presented
Ex: domine = university.

Identify concepts
identify the mean concepts or class in the domine
Ex:- concepts = student, teacher, course, subject

define properity
define attribute for each concepts.
Ex: student: S.name, S.age, S.no, S.city.
teacher: T.name, T.ID, T.city.

define relationship
It tells how the concepts are related to each other
Ex: student studies course

define rules
This are the logical rules or ~~concept~~ constraint.
Ex:- If person studies a course -> student
Every course is offered by a depertement

Create individules
Creates specific instance of concepts
Ex:- student -> Akash
teacher -> Some
course -> BCA

use for reasing

finally ontology use this to derive new information from existing facts and rules.

Ex: Akash belongs to cs department.
if system has facts
Akash studies BCA
BCA is offered by cs department.

ontology tells the computer what concepts at the relationship

semantic web

It is the extance of existing web that allows computer to understand meaning of the data not just display it.

-> semantic web conects the data in meaningful way, so that computer can identify relationship & perform reasing.

purpose of semantic web
-> Machine understanding
-> Data integration.
-> Better searching
-> reasing. (derive new destion.)

- main component
-> URI (Uniform resource indentifer)
-> RDF (Resoree discription framework)
-> ontology.
-> spark QL
-> reasing

-> Suppose we have Akash studies at shrinivas university.
shrinivas university is loaked at mangaluru.

URI
It uniquly identifies each resoree

Ex: ~~step~~

steps in sematic web

step1 :

step2 : information is reporesented ~~using RDF~~ in the form of triples.

step3 : ontology helps to identify meaning and relationship and concepts.
Ex: student, university, city.
every BCA student is a student.

step4 : SparkQL use to ask question about to RDF
ex: where is Akash's university is located.

step5 : reasing derives a new knowledge
reasing allows the system to obtain information that is not directly stated.

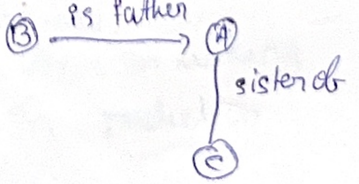
- applications
-> search ingines
-> e-commors
-> health care Ex:- fever ——> symptoms ——> flwo
flwo ——> treated by -> medicine X
-> education
-> recomandation system
-> social network.

Knowledge representation using graph / sematic network

It is the method of representing knowledge. (concepts) as nodes and relationship as edges.

Basic structure.
1) Node -> represents the concepts
2) edges -> represents the relationship.

Ex:- family knowledge using graph



Components

lables -> It discribs the meaning of the node and relationship.

properities -> It is the additional information about the nodes and edges.

advantages

-> Easy to ~~represents~~ represents the relationships
-> helps in reasing
-> useful for searching related information.
-> supports knowledge graph and samatic application

~~Concepti~~

Knowledge represention using graph represents knowledge as inter conected node and relationships allowing AI system to store and reasion about the information.

applications

-> sematic webs
-> chat boat
-> recomendation system
-> search ingines

Representation of knowledge using frames.

frames -> They are knowledge represention technique in AI use to represende object, concepts or suition along with its properities and relaked information.

Note:-
frame is sinciler to record that contains slots and there values.

Ex:- ~~frame~~ = ~~stu~~ student.

slot	value
Name	Akash
age	21
coure	BCA
Roll No	04

Components of frame

1) frame name -> It is identify the concepts.
2) slot -> properities of the concepts.
3) slot value -> It provides actual information.
4) default value -> values are assium when specific information is un cavilable.

advantages

-> simple structure
-> easy to understand.
-> easy to modify.
-> reduces repatation of the information.
-> helps AI system orgnize rulated knowledge

applications

-> robotics
-> NLP
-> computer vision
-> expert system
-> AI reasing systems.

uncertainty in AI

It is the lack of complete knowledge about the currect state outcome. disicision in given suition.

~~cause~~ causes of uncertainity.

-> Incomplete information
-> Noise data -> Data may contain error or incurrent values
-> ambiguous information.
-> lack of knowledge
-> unpredictable events.

effects of uncertainity

-> Incurreet ~~decision~~ decision.
-> redeee accurce.
-> risk in decision making
-> deficulty in predection

technique to handle uncertainity in AI

-> AI handle uncertainity using probability, evidence and reasing techique to make the best possible decision than information incomplete and unclear.

1. probality
2. Fuzzy logic
3. Bayesian Reasoning

* Bayesian ~~Learn~~ theorem

It is methamatical method used in AI and probality to calculate probality of an event based an new or aviable evidence.

-> It is mainly used to uncertainity

Formula

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

P(A|B) : probality of A when B is known
P(B|A) : probality of B when A is true
P(A) : probality of A
P(B) : probality of B (evidence)

Ex suppose a diseas affects 5% of the people if a person as diseas the test is positive 90% of time the test is positive for all 5% of people Find the probality person as diseas when test is positive

-> P(A) = 0.01 -> diseas
P(B) = 0.05 -> test
P(B|A) = 0.9

$$P(A|B) = \frac{0.9 \times 0.01}{0.05}$$

$$P(A|B) = 0.18$$

$$= 18\%$$

2) A student can be good(g) or poor(p) in studies
probality that student is good = 60%
probality that student is poor = 40%.
probality of passing if good = 90%.
probality of passing if poor = 30%
find the probality that student is good even that student as pass

-> $P(G|pass) = \frac{P(pass|G) \times P(G)}{P(pass)}$

P(G) = 60% or 0.6
P(pass|G) = 0.9

P(pass) = P(pass|G) P(G) + P(pass|P) P(P)
= (0.9)(0.6) + (0.3)(0.4)
P(pass) = 0.66

$$P(G|pass) = \frac{(0.9)(0.6)}{(0.66)}$$

$$= \frac{0.54}{0.66}$$

$0.81 = 81\%$

3) probality rain = 30% if trains probality of cold is sky : 80% overall probality is sky = 50%. today sky is colding find the probality. if that will Rain.

-> $P(R|C) = \frac{P(C|R) \times P(R)}{P(C)}$

P(R) = 30%
P(C) = 50%
P(C|R) = 0.8

$$= \frac{(0.8)(0.3)}{(0.5)}$$

$$= 0.48$$

$$= 48\%$$

1) there are two bags bag A
bag b one red balls 4 blue balls.
suppose a bag is selected randomly · P(A) = P(B) = 50%.
A red ball is picked
find the probality come from bag A

→ $P(A|Red) = \frac{P(Red|A) \times P(A)}{P(Red)}$

$= \frac{0.6 \times 0.5}{0.8}$

$= 0.675$

P(A|Red) = 75%.

$P(Red|A) = \frac{3}{5} = 0.6$

$P(Red|B) = \frac{1}{5} = 0.2$

$P(Red) = [P(Red|A) \times P(A)] + [P(Red|B) \times P(B)]$

$= 0.6 \times 0.5 + 0.2 \times 0.5$

$= 0.4$

Fuzzy logic

It is a techique to handle uncertainity in AI

→ It allows values to belong from 0 to 1 means it allows a value to belong to a set partialy

components of Fuzzy logic

1) Crisp input
2) fuzzification
3) rule based → It contains if then rule
4) inference Engine → To findout the new information
5) defuzzification
6) output

Fuzzification

convert the neramical input into fuzzy value such as low, medium, high.

defuzzification

convert the fuzzy result into numarical output

Ex problem statement

An automatic must ajeust it's speed according to room temperature

low temperature slow fan
medeam temperature medium fan
high temperature fast ~~high~~ fan

Find out the ~~speed~~ of the fan the room temperature is 30°

→ step1 = input room temperature 30°, Speed of the fan at 30° temperature.

step2 = fuzzification

temp	numerical value	RPM
low	0.0	400
med	0.4	600
high	0.6	1000

step3: apply fuzzy if than rules
if temperature is low than fan is slow
if temperature is medium than fan is medium
if temperature is high than fan is ~~high~~ fast

step 4: inference

Rules	membership value	Result
low → slow	0.0	no activation
med → med	0.4	moderate
high → fast	0.6	fast.

step 5: defuzzification

$speed = \frac{\Sigma (membership \times RPM)}{\Sigma membership}$

$= \frac{(0.0 \times 400) + (0.4 \times 600) + (0.6 \times 1000)}{0.0 + 0.4 + 0.6}$

$= \frac{0 + 240 + 600}{1}$

$= \frac{840}{1}$

$= 840$ RPM

2) automatic washing machine uses fuzzy logic to determined washing time based on dust level of the clothes.
The membership values & washing time as given below

dust level	membership value	washing time
low	0.2	20
med	0.5	40
high	0.3	60

calculate the final washing time using fuzzy logic

$$FL = \frac{\Sigma \text{membership} \times \text{corresponding o/p}}{\Sigma \text{membership}}$$

$$= \frac{0.2 \times 20 + 0.5 \times 40 + 0.3 \times 60}{0.2 + 0.5 + 0.3}$$

$$= \frac{42}{1}$$

$$= 42$$

→ An automatic Airconditioner uses fuzzy to calculate cooling output based on room temperature

temp	membership value	cooling o/p
low	0.2	200
med	0.3	500
high	0.5	900

$$\rightarrow FL = \frac{\Sigma \text{membership} \times \text{corresponding o/p}}{\Sigma \text{membership}}$$

$$= \frac{0.2 \times 200 + 0.3 \times 500 + 0.5 \times 900}{0.2 + 0.3 + 0.5}$$

$$= \frac{640}{1}$$

$$= 640$$

applications

→ Home ~~applications~~ appliances
→ auto mobile system
→ medical systems
→ robotics
→ trapic control
→ decision making system
→ whether for